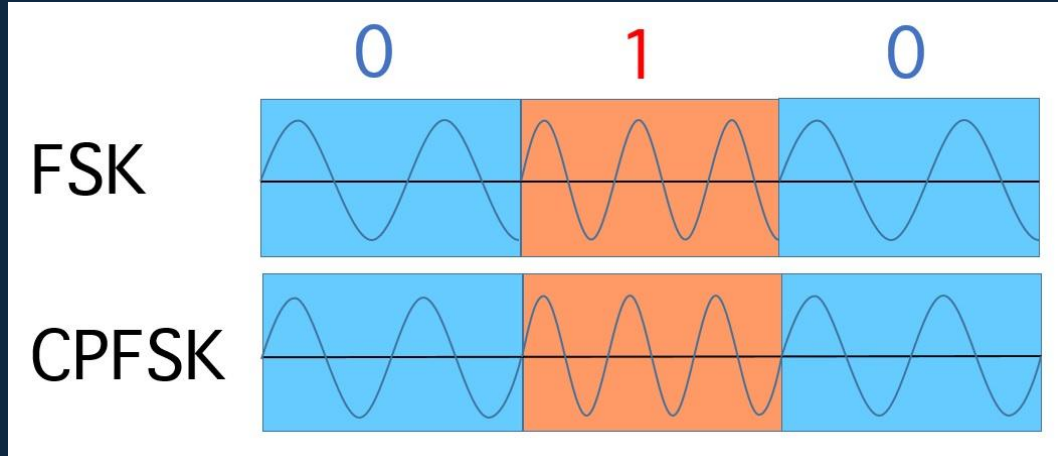


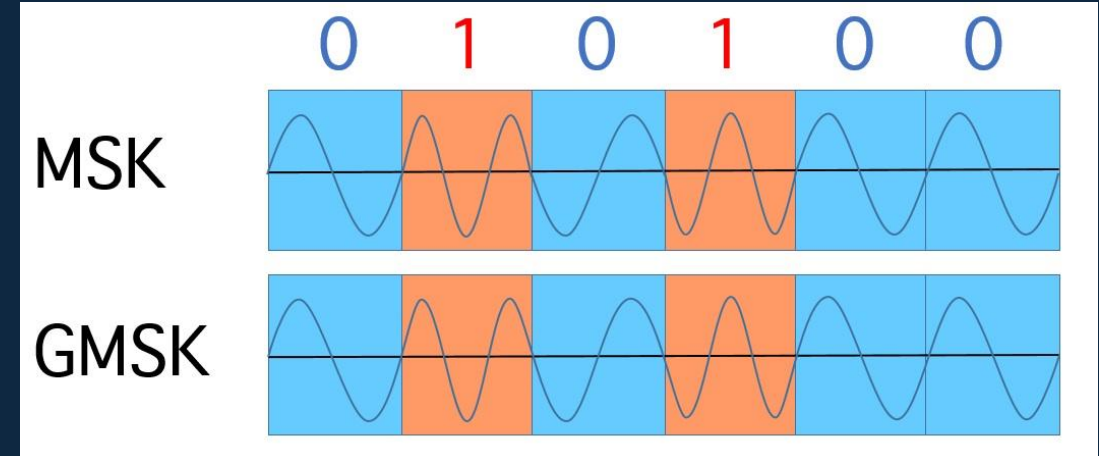
ECE 298A: MSK Modem

Modulation: Varying of one or more properties (amplitude, frequency, phase) of a periodic waveform to transmit information

Minimum-shift keying (MSK): A special case of frequency-shift keying where the phase is continuous (CPFSK) and changes by at most 90° at once



Images by Von Alderamin from scienceblogs.de



Advantages: spectrally efficient compared to regular FSK, resilient to non-linear distortion

Applications: mobile communications (e.g., GSM cell standard), satellite communication

Preliminary Architecture

Will this fit on 1 tile? Probably not...

Will first implement SPI to be able to transmit/receive data to/from chip, synthesize to see area usage, then decide how much to cut off (likely can only keep one of modulator or demodulator)

Specifications + Timeline

Spec	
Symbol rate	200 kbps
Clock	30 MHz
Bit error rate (Receive)	< 0.1% at an Eb/N0 of 10dB
Modulator Input	Serial 1 bit binary, SPI
Demodulator input	4 bit I/Q (total 8 bits), transmitted serially with SPI
Output	4 bit I/Q (total 8 bits), transmitted serially with SPI
Sample rate	8 samples per symbol
Modulator Latency	6 clock cycles - 0.2 us
Demodulator Latency	10 clock cycles - 0.33 us

